

“Digital Forensics”

Let's Grow Smart...

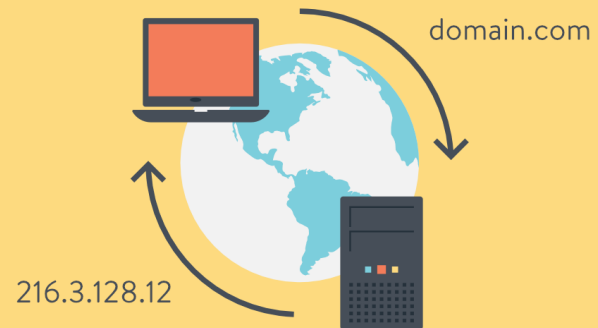
“Introduction”

Let's Learn...

Problem Statement ?

“What is Forensics?”

What to do if we need to find out who committed the crime digitally?

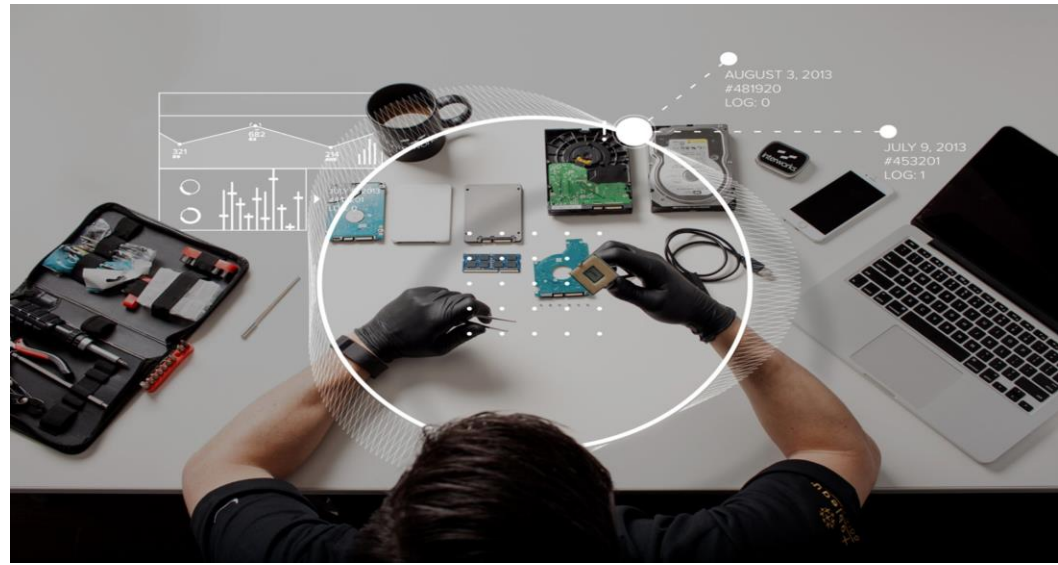


Introduction



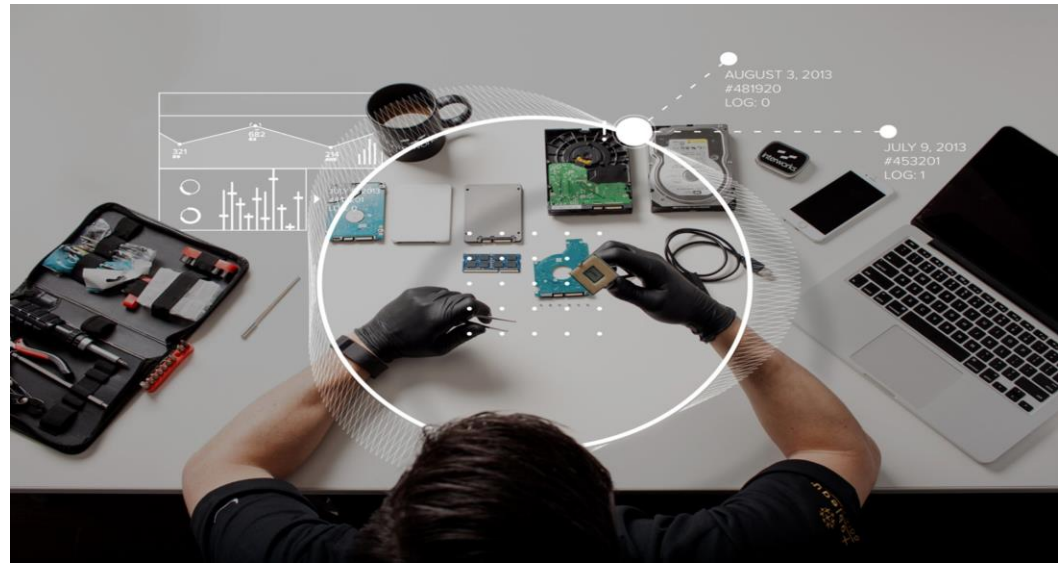
- ▶ “The application of computer science and investigative procedures for a legal purpose involving the analysis of digital evidence after proper search authority, chain of custody, validation with mathematics, use of validated tools, repeat ability, reporting, and possible expert presentation.” (Zatyko, 2007)

Introduction



- ▶ Digital Forensics is defined as the process of preservation, identification, extraction, and documentation of computer evidence which can be used by the court of law.
- ▶ It is a science of finding evidence from digital media like a computer, mobile phone, server, or network.

Introduction



- ▶ Digital forensics encompasses much more than just laptop and desktop computers. Mobile devices, networks, and “cloud” systems are very much within the scope of the discipline.
- ▶ It also includes the analysis of images, videos, and audio (in both analog and digital format). The focus of this kind of analysis is generally authenticity, comparison, and enhancement

Introduction



Objectives of Computer Forensics



- ▶ It helps to recover, analyze, and preserve computer and related materials in such a manner that it helps the investigation agency to present them as evidence in a court of law.
- ▶ It helps to postulate the motive behind the crime and identity of the main culprit.
- ▶ Designing procedures at a suspected crime scene which helps you to ensure that the digital evidence obtained is not corrupted.

Objectives of Computer Forensics

- ▶ Data acquisition and duplication: Recovering deleted files and deleted partitions from digital media to extract the evidence and validate them.
- ▶ Helps you to identify the evidence quickly, and also allows you to estimate the potential impact of the malicious activity on the victim.
- ▶ Producing a computer forensic report which offers a complete report on the investigation process.
- ▶ Preserving the evidence by following the chain of custody.

Types of Digital Forensics

Types of Digital Forensics

Disk Forensics:

It deals with extracting data from storage media by searching active, modified, or deleted files.

Network Forensics:

It is a sub-branch of digital forensics. It is related to monitoring and analysis of computer network traffic to collect important information and legal evidence.

Types of Digital Forensics

Types of Digital Forensics

Wireless Forensics:

It is a division of network forensics. The main aim of wireless forensics is to offer the tools needed to collect and analyze the data from wireless network traffic.

Database Forensics:

It is a branch of digital forensics relating to the study and examination of databases and their related metadata.

Types of Digital Forensics

Types of Digital Forensics

Malware Forensics:

This branch deals with the identification of malicious code, to study their payload, viruses, worms, etc.

Email Forensics

Deals with recovery and analysis of emails, including deleted emails, calendars, and contacts.

Types of Digital Forensics

Types of Digital Forensics

Memory Forensics:

It deals with collecting data from system memory (system registers, cache, RAM) in raw form and then carving the data from Raw dump.

Mobile Phone Forensics:

It mainly deals with the examination and analysis of mobile devices. It helps to retrieve phone and SIM contacts, call logs, incoming, and outgoing SMS/MMS, Audio, videos, etc.

Advantages of Digital Forensics

Advantages of Digital Forensics

- ▶ To ensure the integrity of the computer system.
- ▶ To produce evidence in the court, which can lead to the punishment of the culprit.
- ▶ It helps the companies to capture important information if their computer systems or networks are compromised.
- ▶ Efficiently tracks down cybercriminals from anywhere in the world.
- ▶ Helps to protect the organization's money and valuable time.
- ▶ Allows to extract, process, and interpret the factual evidence, so it proves the cybercriminal action's in the court.

Disadvantages of Digital Forensics

Disadvantages of Digital Forensics

- ▶ Digital evidence accepted into court. However, it is must be proved that there is no tampering.
- ▶ Producing electronic records and storing them is an extremely costly affair.
- ▶ Legal practitioners must have extensive computer knowledge.
- ▶ Need to produce authentic and convincing evidence.
- ▶ If the tool used for digital forensic is not according to specified standards, then in the court of law, the evidence can be disapproved by justice.
- ▶ Lack of technical knowledge by the investigating officer might not offer the desired result.

Process of Digital Forensics

- ▶ Digital forensics entails the following steps:

Identification

Preservation

Analysis

Documentation

Presentation

***Process of
Digital
Forensics***

Process of Digital Forensics

- ▶ Digital forensics entails the following steps:

Identification

- Identify the purpose of investigation
- Identify the resources required

Preservation

- Data is isolate, secure and preserve

Analysis

- Identify tool and techniques to use
- Process data
- Interpret analysis results

Documentation

- Documentation of the crime scene along with photographing, sketching, and crime-scene mapping

Presentation

- Process of summarization and explanation of conclusions is done with the help to gather facts.

Process of Digital Forensics

Any Questions?





THANKS!